

01 — GAN 2D Normalized

Dự án: Latent Manipulation of Brain MRI using Volume-Preserving GANs

Input: PNG normalized từ 00a_preprocessing_2d.ipynb

Kiến trúc:

- **Generator:** U-Net + Age Embedding inject vào bottleneck
- **Discriminator:** PatchGAN + Age Conditioning

Output:

```
gan2d_normalized/  
└─ best_model.pth ← checkpoint tốt nhất (loss_G thấp nhất)
```

Bước 1: Cấu hình

```
import os  
  
# ==== ĐƯỜNG DẪN ====  
DATA_DIR =  
'/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/normalized'  
LABELS_CSV =  
'/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/preprocessing_log.csv'  
OUTPUT_DIR = '/kaggle/working/conditional_gan2d_normalized'  
  
os.makedirs(OUTPUT_DIR, exist_ok=True)  
  
# ==== HYPERPARAMETERS ====  
IMAGE_SIZE = 256  
BATCH_SIZE = 16  
NUM_EPOCHS = 500 # tăng lên 500 cho 2D  
PATIENCE = 20 # dừng nếu val_SSIM không tăng >= MIN_DELTA sau 20 epoch liên tiếp  
LR_G = 2e-4  
LR_D = 2e-4  
LAMBDA_L1 = 10 # giảm từ 100 → 10 để model dám thay đổi tuổi  
LAMBDA_AGE = 5 # tăng từ 1 → 5 để age loss có trọng lượng hơn  
LAMBDA_GP = 10 # gradient penalty cho WGAN-GP  
LATENT_DIM = 256  
  
print(f'Data dir : {DATA_DIR}')
```

```
print(f'PNG files: {len([f for f in os.listdir(DATA_DIR) if  
f.endswith(".png")])})')
```

```
Data dir :  
/kaggle/input/datasets/minhbodoi/full-preprocessed-2d/preprocessed_2d/  
normalized  
PNG files: 12497
```

Bước 2: Import thư viện

```
import torch  
import torch.nn as nn  
import torch.optim as optim  
from torch.utils.data import Dataset, DataLoader  
import torchvision.transforms as transforms  
import numpy as np  
import pandas as pd  
from PIL import Image  
from tqdm.notebook import tqdm  
  
DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'  
print(f'Device: {DEVICE}')if DEVICE == 'cuda':  
    print(f'GPU : {torch.cuda.get_device_name(0)}')    print(f'VRAM: {torch.cuda.get_device_properties(0).total_memory /  
1e9:.1f} GB')  
Device: cuda  
GPU : Tesla T4  
VRAM: 15.6 GB
```

Bước 3: Dataset

```
from torch.utils.data import random_split  
  
class BrainMRI2DDataset(Dataset):  
    def __init__(self, data_dir, labels_csv, image_size=256):  
        self.data_dir = data_dir  
        df = pd.read_csv(labels_csv)  
        df['full_path'] = df['png_file'].apply(lambda x:  
os.path.join(data_dir, x))  
        df =  
df[df['full_path'].apply(os.path.exists)].reset_index(drop=True)  
        self.df = df  
        self.age_min = df['age'].min()  
        self.age_max = df['age'].max()  
        self.transform = transforms.Compose([  
            transforms.Resize((image_size, image_size)),  
            transforms.ToTensor(),  
            transforms.Normalize([0.5], [0.5])  
        ])  
        print(f'Dataset: {len(df)} ảnh | tuổi [{self.age_min:.1f},
```

```

{self.age_max:.1f}']')

    def normalize_age(self, age):
        return 2 * (age - self.age_min) / (self.age_max -
self.age_min) - 1

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        img = Image.open(row['full_path']).convert('L')
        img = self.transform(img)
        age_norm = torch.tensor(self.normalize_age(row['age']),
dtype=torch.float32)
        age_raw = torch.tensor(row['age'] / 100.0,
dtype=torch.float32)
        return img, age_norm, age_raw

# — Train / Test split 80/20 —————
full_dataset = BrainMRI2DDataset(DATA_DIR, LABELS_CSV, IMAGE_SIZE)
n_total = len(full_dataset)
n_train = int(0.8 * n_total)
n_test = n_total - n_train

train_set, test_set = random_split(
    full_dataset, [n_train, n_test],
    generator=torch.Generator().manual_seed(42) # seed cố định để
tái lập
)

# Lưu lại age_min/max từ full dataset (không tính riêng train)
dataset = full_dataset # dùng để lấy age_min/max khi save
checkpoint

dataloader = DataLoader(train_set, batch_size=BATCH_SIZE,
                        shuffle=True, num_workers=4,
pin_memory=True)
dataloader_test = DataLoader(test_set, batch_size=BATCH_SIZE,
                             shuffle=False, num_workers=4,
pin_memory=True)

print(f'Train: {n_train} ảnh | Test: {n_test} ảnh')

Dataset: 12497 ảnh | tuổi [18.0, 97.0]
Train: 9997 ảnh | Test: 2500 ảnh

```

Bước 4: Kiến trúc Model

```
class AgeEmbedding(nn.Module):
    def __init__(self, embed_dim=256):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(1, 128), nn.ReLU(),
            nn.Linear(128, embed_dim)
        )
    def forward(self, age):
        return self.net(age.unsqueeze(-1))

class UNetBlock(nn.Module):
    def __init__(self, in_ch, out_ch, down=True, use_bn=True,
        dropout=False):
        super().__init__()
        layers = []
        if down : layers.append(nn.Conv2d(in_ch, out_ch, 4, 2, 1,
            bias=False))
        else : layers.append(nn.ConvTranspose2d(in_ch, out_ch, 4,
            2, 1, bias=False))
        if use_bn : layers.append(nn.BatchNorm2d(out_ch))
        if dropout : layers.append(nn.Dropout(0.5))
        layers.append(nn.LeakyReLU(0.2) if down else nn.ReLU())
        self.block = nn.Sequential(*layers)
    def forward(self, x):
        return self.block(x)

class Generator(nn.Module):
    """
    U-Net Generator với age conditioning.
    Input : ảnh (B, 1, 256, 256) + tuổi normalize (B,)
    Output: ảnh sinh ra (B, 1, 256, 256)
    """
    def __init__(self, latent_dim=256):
        super().__init__()
        self.age_embed = AgeEmbedding(latent_dim)
        self.age_proj = nn.Linear(latent_dim, 512)
        self.age_fuse = nn.Conv2d(1024, 512, 1) # concat e8(512) +
age(512) → 512
        self.e1 = UNetBlock(1, 64, down=True, use_bn=False)
        self.e2 = UNetBlock(64, 128, down=True)
        self.e3 = UNetBlock(128, 256, down=True)
        self.e4 = UNetBlock(256, 512, down=True)
        self.e5 = UNetBlock(512, 512, down=True)
        self.e6 = UNetBlock(512, 512, down=True)
        self.e7 = UNetBlock(512, 512, down=True)
        self.e8 = UNetBlock(512, 512, down=True, use_bn=False)
```

```

        self.d1 = UNetBlock(512, 512, down=False, dropout=True)
        self.d2 = UNetBlock(1024, 512, down=False, dropout=True)
        self.d3 = UNetBlock(1024, 512, down=False, dropout=True)
        self.d4 = UNetBlock(1024, 512, down=False)
        self.d5 = UNetBlock(1024, 256, down=False)
        self.d6 = UNetBlock(512, 128, down=False)
        self.d7 = UNetBlock(256, 64, down=False)
        self.out = nn.Sequential(nn.ConvTranspose2d(128, 1, 4, 2, 1),
nn.Tanh())

    def forward(self, x, age):
        e1 = self.e1(x); e2 = self.e2(e1); e3 = self.e3(e2); e4 =
self.e4(e3)
        e5 = self.e5(e4); e6 = self.e6(e5); e7 = self.e7(e6); e8 =
self.e8(e7)
        age_feat = self.age_proj(self.age_embed(age)).view(-1, 512, 1,
1).expand_as(e8)
        z = self.age_fuse(torch.cat([e8, age_feat], dim=1)) # concat
→ model tự học balance
        d1 = self.d1(z)
        d2 = self.d2(torch.cat([d1, e7], dim=1))
        d3 = self.d3(torch.cat([d2, e6], dim=1))
        d4 = self.d4(torch.cat([d3, e5], dim=1))
        d5 = self.d5(torch.cat([d4, e4], dim=1))
        d6 = self.d6(torch.cat([d5, e3], dim=1))
        d7 = self.d7(torch.cat([d6, e2], dim=1))
        return self.out(torch.cat([d7, e1], dim=1))

class Discriminator(nn.Module):
    """
    PatchGAN Discriminator với age conditioning.
    Input : ảnh (B, 1, H, W) + age channel → (B, 2, H, W)
    """
    def __init__(self):
        super().__init__()
        self.model = nn.Sequential(
            nn.Conv2d(2, 64, 4, 2, 1, bias=False), nn.LeakyReLU(0.2),
            nn.Conv2d(64, 128, 4, 2, 1, bias=False),
nn.BatchNorm2d(128), nn.LeakyReLU(0.2),
            nn.Conv2d(128, 256, 4, 2, 1, bias=False),
nn.BatchNorm2d(256), nn.LeakyReLU(0.2),
            nn.Conv2d(256, 512, 4, 1, 1, bias=False),
nn.BatchNorm2d(512), nn.LeakyReLU(0.2),
            nn.Conv2d(512, 1, 4, 1, 1)
        )
    def forward(self, img, age):
        age_map = age.view(-1, 1, 1, 1).expand(-1, 1, img.shape[2],
img.shape[3])
        return self.model(torch.cat([img, age_map], dim=1))

```

```

G = Generator(LATENT_DIM).to(DEVICE)
D = Discriminator().to(DEVICE)
print(f'Generator      : {sum(p.numel() for p in
G.parameters())/1e6:.1f}M params')
print(f'Discriminator: {sum(p.numel() for p in
D.parameters())/1e6:.1f}M params')

Generator      : 55.1M params
Discriminator: 2.8M params

```

Bước 5: Loss + Optimizers

Dùng **WGAN-GP** thay vì BCE để tránh mode collapse và training ổn định hơn.

```

# — WGAN-GP: không dùng BCEWithLogitsLoss nữa —————
# Discriminator không dùng sigmoid, output là raw score (Wasserstein)
criterion_l1 = nn.L1Loss()
criterion_age = nn.MSELoss()

def compute_gradient_penalty(D, real, fake, ages, device):
    """Gradient penalty để ổn định training WGAN-GP."""
    B = real.size(0)
    alpha = torch.rand(B, 1, 1, 1, device=device)
    interp = (alpha * real + (1 - alpha) * fake).requires_grad_(True)
    d_interp = D(interp, ages)
    grad = torch.autograd.grad(
        outputs=d_interp, inputs=interp,
        grad_outputs=torch.ones_like(d_interp),
        create_graph=True, retain_graph=True
    )[0]
    gp = ((grad.norm(2, dim=1) - 1) ** 2).mean()
    return gp

age_regressor = nn.Sequential(
    nn.AdaptiveAvgPool2d(8), nn.Flatten(),
    nn.Linear(64, 64), nn.ReLU(),
    nn.Linear(64, 1), nn.Sigmoid()
).to(DEVICE)

opt_G = optim.Adam(
    list(G.parameters()) + list(age_regressor.parameters()),
    lr=LR_G, betas=(0.0, 0.9) # betas WGAN-GP chuẩn
)
opt_D = optim.Adam(D.parameters(), lr=LR_D, betas=(0.0, 0.9))

scheduler_G = optim.lr_scheduler.StepLR(opt_G, step_size=50,
gamma=0.5)

```

```
scheduler_D = optim.lr_scheduler.StepLR(opt_D, step_size=50,  
gamma=0.5)
```

```
print('Loss + Optimizers (WGAN-GP) sẵn sàng!')
```

```
Loss + Optimizers (WGAN-GP) sẵn sàng!
```

Bước 6: Training

```
from skimage.metrics import structural_similarity as ssim_fn  
import numpy as np, subprocess, shutil, tempfile  
  
# — Duong dan checkpoint —————  
CKPT_PATH      = f'{OUTPUT_DIR}/last_checkpoint.pth'  
BEST_PATH      = f'{OUTPUT_DIR}/best_model.pth'  
  
# — Kaggle Dataset info —————  
_ku = subprocess.run('kaggle config view', shell=True,  
capture_output=True, text=True).stdout  
KAGGLE_USER    = [l.split(':')[1].strip() for l in _ku.split('\n') if  
'username' in l][0]  
DATASET_NAME   = os.path.basename(OUTPUT_DIR).replace('_', '-')  
MIN_DELTA      = 0.005 # chỉ tính là 'cải thiện thật' khi SSIM tăng >=  
0.005  
  
def push_checkpoints_to_kaggle(msg):  
    """Push chỉ 2 file .pth lên Kaggle Dataset (version mới để lên  
version cũ)."""  
    tmp = tempfile.mkdtemp()  
    try:  
        for fn in ['last_checkpoint.pth', 'best_model.pth']:  
            if os.path.exists(f'{OUTPUT_DIR}/{fn}'):   
                shutil.copy2(f'{OUTPUT_DIR}/{fn}', f'{tmp}/{fn}')  
        import json as _j  
        with open(f'{tmp}/dataset-metadata.json', 'w') as _f:  
            _j.dump({'title': DATASET_NAME,  
                    'id'    : f'{KAGGLE_USER}/{DATASET_NAME}',  
                    'licenses': [{ 'name': 'CC0-1.0' }]}, _f)  
        chk = subprocess.run(  
            f'kaggle datasets list --user {KAGGLE_USER} --search  
{DATASET_NAME}',  
            shell=True, capture_output=True, text=True)  
        if DATASET_NAME in chk.stdout:  
            subprocess.run(f'kaggle datasets version -p {tmp} -m  
"{msg}"', shell=True)  
        else:  
            subprocess.run(f'kaggle datasets create -p {tmp}',  
shell=True)  
        print(f' Cloud: {msg} -> {KAGGLE_USER}/{DATASET_NAME}')  
    finally:
```

```

        shutil.rmtree(tmp)

# — Resume neu co checkpoint —————
start_epoch      = 1
patience_count  = 0
best_val_ssim    = -1.0
history = {'loss_G': [], 'loss_D': [], 'loss_L1': [], 'loss_age': [],
          'val_ssim': []}

# Tim checkpoint: uu tien /kaggle/working, neu khong co thi tim
/kaggle/input
def find_checkpoint(filename):
    # 1. Working dir (session hien tai)
    p = f'{OUTPUT_DIR}/{filename}'
    if os.path.exists(p): return p, 'working'
    # 2. Kaggle Dataset input (push tu session truoc)
    import glob
    matches = glob.glob(f'/kaggle/input/**/{filename}',
recursive=True)
    if matches: return matches[0], 'input'
    return None, None

load_path = None
for fname in ['last_checkpoint.pth', 'best_model.pth']:
    p, src_loc = find_checkpoint(fname)
    if p:
        load_path = p
        print(f'Tim thay {fname} ({src_loc}) – load de train tiep...')
        break
if not load_path:
    print('Khong co checkpoint – train tu dau')

if load_path:
    ckpt = torch.load(load_path, map_location=DEVICE)
    G.load_state_dict(ckpt['G_state'])
    D.load_state_dict(ckpt['D_state'])
    opt_G.load_state_dict(ckpt['opt_G'])
    opt_D.load_state_dict(ckpt['opt_D'])
    start_epoch      = ckpt['epoch'] + 1
    best_val_ssim     = ckpt.get('best_val_ssim', -1.0)
    patience_count    = ckpt.get('patience_count', 0)
    history           = ckpt.get('history', history)
    print(f'Resume tu epoch {ckpt["epoch"]} |
best_val_SSIM={best_val_ssim:.4f}')
    print(f'    Tiep tuc tu epoch {start_epoch} -> {NUM_EPOCHS}')

N_CRITIC = 5
print(f'Bat dau training {NUM_EPOCHS} epochs (WGAN-GP)...\n')

for epoch in range(start_epoch, NUM_EPOCHS + 1):

```

```

G.train(); D.train()
epoch_loss_G = epoch_loss_D = epoch_loss_L1 = epoch_loss_age = 0
n_batches = 0

data_iter = iter(dataloader)
for _ in range(len(dataloader) // N_CRITIC):
    for _ in range(N_CRITIC):
        try:
            real_imgs, ages_norm, ages_raw = next(data_iter)
        except StopIteration:
            data_iter = iter(dataloader)
            real_imgs, ages_norm, ages_raw = next(data_iter)
        real_imgs = real_imgs.to(DEVICE)
        ages_norm = ages_norm.to(DEVICE)
        ages_raw = ages_raw.to(DEVICE)
        opt_D.zero_grad()
        with torch.no_grad():
            fake_imgs = G(real_imgs, ages_norm)
        loss_D_real = -D(real_imgs, ages_norm).mean()
        loss_D_fake = D(fake_imgs.detach(), ages_norm).mean()
        gp = compute_gradient_penalty(D, real_imgs,
fake_imgs.detach(), ages_norm, DEVICE)
        loss_D = loss_D_real + loss_D_fake + LAMBDA_GP * gp
        loss_D.backward()
        opt_D.step()
        epoch_loss_D += loss_D.item()

    try:
        real_imgs, ages_norm, ages_raw = next(data_iter)
    except StopIteration:
        data_iter = iter(dataloader)
        real_imgs, ages_norm, ages_raw = next(data_iter)
    real_imgs = real_imgs.to(DEVICE)
    ages_norm = ages_norm.to(DEVICE)
    ages_raw = ages_raw.to(DEVICE)
    opt_G.zero_grad()
    fake_imgs = G(real_imgs, ages_norm)
    loss_G_adv = -D(fake_imgs, ages_norm).mean()
    loss_L1 = criterion_l1(fake_imgs, real_imgs) * LAMBDA_L1
    pred_age = age_regressor(fake_imgs).squeeze()
    loss_age = criterion_age(pred_age, ages_raw) * LAMBDA_AGE
    loss_G = loss_G_adv + loss_L1 + loss_age
    loss_G.backward()
    opt_G.step()
    epoch_loss_G += loss_G_adv.item()
    epoch_loss_L1 += loss_L1.item()
    epoch_loss_age += loss_age.item()
    n_batches += 1

scheduler_G.step()

```

```

scheduler_D.step()

n = max(n_batches, 1)
avg_loss_G = epoch_loss_G / n
avg_loss_D = epoch_loss_D / (n * N_CRITIC)
avg_loss_L1 = epoch_loss_L1 / n
avg_loss_age = epoch_loss_age / n

# — SSIM tren test set voi shuffled age —————
G.eval()
ssim_scores = []
with torch.no_grad():
    for real_imgs, ages_norm, _ in dataloader_test:
        real_imgs = real_imgs.to(DEVICE)
        shuffled_idx = torch.randperm(len(ages_norm))
        ages_shuffled = ages_norm[shuffled_idx].to(DEVICE)
        fake_imgs = G(real_imgs, ages_shuffled)
        for i in range(real_imgs.size(0)):
            r = (real_imgs[i, 0].cpu().numpy() + 1) / 2
            f = (fake_imgs[i, 0].cpu().numpy() + 1) / 2
            ssim_scores.append(ssim_fn(r, f, data_range=1.0))
val_ssim = float(np.mean(ssim_scores))

history['loss_G'].append(avg_loss_G)
history['loss_D'].append(avg_loss_D)
history['loss_L1'].append(avg_loss_L1)
history['loss_age'].append(avg_loss_age)
history['val_ssim'].append(val_ssim)

# — Luu last_checkpoint —————
torch.save({
    'epoch' : epoch,
    'G_state' : G.state_dict(),
    'D_state' : D.state_dict(),
    'opt_G' : opt_G.state_dict(),
    'opt_D' : opt_D.state_dict(),
    'history' : history,
    'age_min' : dataset.age_min,
    'age_max' : dataset.age_max,
    'best_val_ssim' : best_val_ssim,
    'patience_count' : patience_count,
    'test_indices' : test_set.indices,
}, CKPT_PATH)

# — Luu best model neu cai thien —————
if val_ssim > best_val_ssim + MIN_DELTA:
    best_val_ssim = val_ssim
    patience_count = 0
    torch.save({
        'epoch' : epoch,

```

```

        'G_state'      : G.state_dict(),
        'D_state'      : D.state_dict(),
        'opt_G'        : opt_G.state_dict(),
        'opt_D'        : opt_D.state_dict(),
        'history'       : history,
        'age_min'       : dataset.age_min,
        'age_max'       : dataset.age_max,
        'best_loss_G'   : avg_loss_G,
        'best_val_ssim' : best_val_ssim,
        'val_ssim'      : val_ssim,
        'test_indices'  : test_set.indices,
    }, BEST_PATH)
    print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
          f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '
          f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} | '
[BEST]')
    else:
        patience_count += 1
        print(f'Epoch {epoch:3d}/{NUM_EPOCHS} | '
              f'loss_G={avg_loss_G:.4f} | loss_D={avg_loss_D:.4f} | '
              f'loss_L1={avg_loss_L1:.4f} | val_SSIM={val_ssim:.4f} | '
              f'no improve {patience_count}/{PATIENCE}')

# — Push moi epoch (last_checkpoint + best_model) —————
push_checkpoints_to_kaggle(f'{epoch}/{NUM_EPOCHS}')

# — Early stopping —————
if patience_count >= PATIENCE:
    print(f'Early stopping tai epoch {epoch}')
    break

print(f'\n=== TRAINING HOAN THANH ===')
print(f'Best val_SSIM: {best_val_ssim:.4f}')
print(f'Model luu tai: {OUTPUT_DIR}/best_model.pth')

```

Không có checkpoint – train từ đầu
 Bắt đầu training 500 epochs (WGAN-GP)...

```

Epoch   1/500 | loss_G=1.8895 | loss_D=0.6143 | loss_L1=1.1392 |
val_SSIM=0.7216 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth
100%|██████████| 662M/662M [00:09<00:00, 71.5MB/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

```

100%|██████████| 662M/662M [00:10<00:00, 67.4MB/s]

Upload successful: last_checkpoint.pth (662MB)

Your private Dataset is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan2d-normalized>

Cloud: 1/500 -> cminhnguyndsdsds/conditional-gan2d-normalized

Epoch 2/500 | loss_G=2.9987 | loss_D=1.5384 | loss_L1=0.3892 | val_SSIM=0.8054 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 87.2MB/s]

0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:12<00:00, 55.7MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan2d-normalized>

Cloud: 2/500 -> cminhnguyndsdsds/conditional-gan2d-normalized

Epoch 3/500 | loss_G=1.5562 | loss_D=1.5446 | loss_L1=0.3241 | val_SSIM=0.8206 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:10<00:00, 67.1MB/s]

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100%|██████████| 662M/662M [00:10<00:00, 68.2MB/s]

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Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdsds/conditional-gan2d-normalized>

Cloud: 3/500 -> cminhnguyndsdsds/conditional-gan2d-normalized

Epoch 4/500 | loss_G=0.4989 | loss_D=1.7776 | loss_L1=0.3120 | val_SSIM=0.8523 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

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100%|██████████| 662M/662M [00:10<00:00, 69.2MB/s]
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100%|██████████| 662M/662M [00:09<00:00, 75.1MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 4/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 5/500 | loss_G=-1.2237 | loss_D=1.8278 | loss_L1=0.3186 |
val_SSIM=0.9486 | [BEST]

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(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:11<00:00, 62.2MB/s]

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100%|██████████| 662M/662M [00:10<00:00, 68.4MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 5/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 6/500 | loss_G=-4.7942 | loss_D=1.9170 | loss_L1=0.3179 |
val_SSIM=0.9381 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:09<00:00, 71.4MB/s]
2%| | 16.4M/662M [00:00<00:03, 172MB/s]

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100%|██████████| 662M/662M [00:09<00:00, 72.1MB/s]

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Dataset version is being created. Please check progress at

<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 6/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 7/500 | loss_G=-7.9741 | loss_D=1.8853 | loss_L1=0.2971 |
val_SSIM=0.9094 | no improve 2/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:09<00:00, 73.7MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 80.9MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 7/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 8/500 | loss_G=-8.6487 | loss_D=1.8940 | loss_L1=0.2851 |
val_SSIM=0.9067 | no improve 3/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:07<00:00, 89.3MB/s]

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100%|██████████| 662M/662M [00:09<00:00, 76.3MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 8/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 9/500 | loss_G=-10.7722 | loss_D=1.8402 | loss_L1=0.2959 |
val_SSIM=0.8874 | no improve 4/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:05<00:00, 128MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 127MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 9/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 10/500 | loss_G=-8.4749 | loss_D=1.9492 | loss_L1=0.2910 |
val_SSIM=0.9385 | no improve 5/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 92.5MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 87.9MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 10/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 11/500 | loss_G=-9.7760 | loss_D=1.8802 | loss_L1=0.3151 |
val_SSIM=0.8989 | no improve 6/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 98.3MB/s]

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100%|██████████| 662M/662M [00:10<00:00, 65.3MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 11/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 12/500 | loss_G=-11.6823 | loss_D=1.9731 | loss_L1=0.3084 |
val_SSIM=0.9437 | no improve 7/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

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100%|██████████| 662M/662M [00:08<00:00, 77.4MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 126MB/s]

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Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 12/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 13/500 | loss_G=-11.1534 | loss_D=2.1274 | loss_L1=0.2972 |
val_SSIM=0.9380 | no improve 8/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 87.9MB/s]
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100%|██████████| 662M/662M [00:06<00:00, 104MB/s]

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Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 13/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 14/500 | loss_G=-10.8697 | loss_D=1.9579 | loss_L1=0.2816 |
val_SSIM=0.9242 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

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100%|██████████| 662M/662M [00:09<00:00, 75.2MB/s]
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100%|██████████| 662M/662M [00:10<00:00, 67.1MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 14/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 15/500 | loss_G=-8.8526 | loss_D=1.9084 | loss_L1=0.2696 |
val_SSIM=0.9364 | no improve 10/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:11<00:00, 58.1MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 92.5MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 15/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 16/500 | loss_G=-9.5309 | loss_D=1.8910 | loss_L1=0.2661 |
val_SSIM=0.9560 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:08<00:00, 84.4MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 125MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 16/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 17/500 | loss_G=-8.4346 | loss_D=1.7535 | loss_L1=0.2489 |
val_SSIM=0.9671 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
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100%|██████████| 662M/662M [00:09<00:00, 71.7MB/s]
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100%|██████████| 662M/662M [00:09<00:00, 76.6MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 17/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 18/500 | loss_G=-9.1451 | loss_D=1.7431 | loss_L1=0.2432 |
val_SSIM=0.8939 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:06<00:00, 115MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 83.2MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 18/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 19/500 | loss_G=-8.9335 | loss_D=1.8488 | loss_L1=0.2498 |
val_SSIM=0.9611 | no improve 2/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 91.6MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 91.5MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 19/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 20/500 | loss_G=-7.2902 | loss_D=1.5771 | loss_L1=0.2106 |
val_SSIM=0.9507 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:06<00:00, 109MB/s]
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100%|██████████| 662M/662M [00:06<00:00, 104MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 20/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 21/500 | loss_G=-6.7426 | loss_D=1.5681 | loss_L1=0.2114 |
val_SSIM=0.9597 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:10<00:00, 68.9MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 88.5MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 21/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 22/500 | loss_G=-8.2690 | loss_D=1.7318 | loss_L1=0.2500 |
val_SSIM=0.9602 | no improve 5/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:09<00:00, 73.6MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 126MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 22/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 23/500 | loss_G=-8.1236 | loss_D=1.4195 | loss_L1=0.2086 |
val_SSIM=0.9691 | no improve 6/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:05<00:00, 132MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 23/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 24/500 | loss_G=-5.4584 | loss_D=1.3903 | loss_L1=0.2076 |
val_SSIM=0.9496 | no improve 7/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 89.1MB/s]
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100%|██████████| 662M/662M [00:05<00:00, 128MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 24/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 25/500 | loss_G=-4.6054 | loss_D=1.1634 | loss_L1=0.1954 |

val_SSIM=0.9705 | no improve 8/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:08<00:00, 79.2MB/s]
0%| | 1.19M/662M [00:00<00:55, 12.4MB/s]

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100%|██████████| 662M/662M [00:08<00:00, 84.5MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 25/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 26/500 | loss_G=-4.5329 | loss_D=1.0647 | loss_L1=0.1850 |
val_SSIM=0.9634 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 96.5MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:04<00:00, 139MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 26/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 27/500 | loss_G=-3.7091 | loss_D=1.0171 | loss_L1=0.1876 |
val_SSIM=0.9618 | no improve 10/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:10<00:00, 67.1MB/s]
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Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:08<00:00, 83.4MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 27/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 28/500 | loss_G=-3.5535 | loss_D=1.0103 | loss_L1=0.1925 |
val_SSIM=0.9670 | no improve 11/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 83.8MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 137MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 28/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 29/500 | loss_G=-3.0545 | loss_D=0.9129 | loss_L1=0.1830 |
val_SSIM=0.9722 | [BEST]
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:09<00:00, 75.9MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:04<00:00, 141MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 29/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 30/500 | loss_G=-2.7258 | loss_D=0.9300 | loss_L1=0.1898 |
val_SSIM=0.9676 | no improve 1/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 85.2MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:10<00:00, 67.1MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 30/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 31/500 | loss_G=-2.1731 | loss_D=0.8907 | loss_L1=0.1851 |
val_SSIM=0.9718 | no improve 2/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:08<00:00, 83.1MB/s]
0%| | 896k/662M [00:00<01:18, 8.82MB/s]

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100%|██████████| 662M/662M [00:07<00:00, 96.5MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 31/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 32/500 | loss_G=-2.1077 | loss_D=0.8733 | loss_L1=0.1832 |
val_SSIM=0.9593 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:08<00:00, 77.8MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 81.3MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 32/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 33/500 | loss_G=-3.2021 | loss_D=0.8297 | loss_L1=0.1875 |
val_SSIM=0.9622 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 82.1MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 77.8MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 33/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 34/500 | loss_G=-2.5982 | loss_D=0.7864 | loss_L1=0.1943 |
val_SSIM=0.9658 | no improve 5/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 81.0MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 82.4MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 34/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 35/500 | loss_G=-2.0041 | loss_D=0.7832 | loss_L1=0.1892 |
val_SSIM=0.9670 | no improve 6/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 119MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 123MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 35/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 36/500 | loss_G=-1.9777 | loss_D=0.7625 | loss_L1=0.1999 |
val_SSIM=0.9609 | no improve 7/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 80.7MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:05<00:00, 119MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 36/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 37/500 | loss_G=-1.2242 | loss_D=0.7811 | loss_L1=0.1924 |
val_SSIM=0.9574 | no improve 8/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)
Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 117MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:05<00:00, 127MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 37/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 38/500 | loss_G=-1.3034 | loss_D=0.7689 | loss_L1=0.1858 |
val_SSIM=0.9620 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 89.5MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 85.7MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 38/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 39/500 | loss_G=-0.2814 | loss_D=0.7712 | loss_L1=0.1867 |
val_SSIM=0.9686 | no improve 10/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:05<00:00, 122MB/s]
1%| | 8.81M/662M [00:00<00:08, 83.3MB/s]

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100%|██████████| 662M/662M [00:06<00:00, 101MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 39/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 40/500 | loss_G=-0.5894 | loss_D=0.7630 | loss_L1=0.1873 |
val_SSIM=0.9725 | no improve 11/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 86.8MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:09<00:00, 70.7MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 40/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 41/500 | loss_G=0.1323 | loss_D=0.7062 | loss_L1=0.1810 |
val_SSIM=0.9755 | no improve 12/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 88.3MB/s]

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100%|██████████| 662M/662M [00:05<00:00, 133MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 41/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 42/500 | loss_G=0.0894 | loss_D=0.7325 | loss_L1=0.1765 |
val_SSIM=0.9663 | no improve 13/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 111MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 79.3MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 42/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 43/500 | loss_G=-0.2491 | loss_D=0.7124 | loss_L1=0.1752 |
val_SSIM=0.9668 | no improve 14/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 94.1MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 87.2MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 43/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 44/500 | loss_G=0.1269 | loss_D=0.6755 | loss_L1=0.1756 | val_SSIM=0.9778 | [BEST]

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 91.9MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:07<00:00, 87.2MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at <https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 44/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 45/500 | loss_G=-0.1593 | loss_D=0.6647 | loss_L1=0.1738 | val_SSIM=0.9733 | no improve 1/20

Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.1), please consider upgrading to the latest version (2.0.2)

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100%|██████████| 662M/662M [00:07<00:00, 96.7MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 83.8MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 45/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 46/500 | loss_G=0.3619 | loss_D=0.6810 | loss_L1=0.1777 |
val_SSIM=0.9722 | no improve 2/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 94.0MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:10<00:00, 65.2MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 46/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 47/500 | loss_G=0.2733 | loss_D=0.6891 | loss_L1=0.1779 |
val_SSIM=0.9737 | no improve 3/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:10<00:00, 67.4MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 92.9MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 47/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 48/500 | loss_G=-0.2034 | loss_D=0.6678 | loss_L1=0.1749 |
val_SSIM=0.9627 | no improve 4/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:06<00:00, 106MB/s]
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100%|██████████| 662M/662M [00:07<00:00, 92.4MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 48/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 49/500 | loss_G=0.1015 | loss_D=0.7138 | loss_L1=0.1836 |
val_SSIM=0.9593 | no improve 5/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 82.1MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 91.8MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 49/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 50/500 | loss_G=0.4473 | loss_D=0.7041 | loss_L1=0.1905 |
val_SSIM=0.9635 | no improve 6/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

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100%|██████████| 662M/662M [00:04<00:00, 140MB/s]
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100%|██████████| 662M/662M [00:08<00:00, 78.4MB/s]

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Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 50/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 51/500 | loss_G=0.2118 | loss_D=0.5863 | loss_L1=0.1749 |
val_SSIM=0.9719 | no improve 7/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 83.1MB/s]
2%|███████| 10.7M/662M [00:00<00:06, 112MB/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 122MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 51/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 52/500 | loss_G=0.3135 | loss_D=0.6147 | loss_L1=0.1782 |
val_SSIM=0.9634 | no improve 8/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 134MB/s]
0%|███████| 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:09<00:00, 71.3MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 52/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 53/500 | loss_G=0.3410 | loss_D=0.6010 | loss_L1=0.1706 |
val_SSIM=0.9687 | no improve 9/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 78.6MB/s]
0%|███████| 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:10<00:00, 65.6MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 53/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 54/500 | loss_G=0.5969 | loss_D=0.6012 | loss_L1=0.1723 |
val_SSIM=0.9679 | no improve 10/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 79.4MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)
Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 87.3MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 54/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 55/500 | loss_G=0.7553 | loss_D=0.6240 | loss_L1=0.1714 |
val_SSIM=0.9628 | no improve 11/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:06<00:00, 104MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 123MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 55/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 56/500 | loss_G=-0.1343 | loss_D=0.5714 | loss_L1=0.1633 |

val_SSIM=0.9742 | no improve 12/20
Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:10<00:00, 65.0MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 87.4MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsd/sds/conditional-gan2d-normalized>

Cloud: 56/500 -> cminhnguyndsd/sds/conditional-gan2d-normalized
Epoch 57/500 | loss_G=0.0617 | loss_D=0.6009 | loss_L1=0.1683 |
val_SSIM=0.9549 | no improve 13/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:09<00:00, 71.9MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:08<00:00, 84.4MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsd/sds/conditional-gan2d-normalized>

Cloud: 57/500 -> cminhnguyndsd/sds/conditional-gan2d-normalized
Epoch 58/500 | loss_G=0.4955 | loss_D=0.6026 | loss_L1=0.1634 |
val_SSIM=0.9764 | no improve 14/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:09<00:00, 73.5MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:07<00:00, 93.9MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 58/500 -> cminhnguyndsdssds/conditional-gan2d-normalized

Epoch 59/500 | loss_G=0.2498 | loss_D=0.5721 | loss_L1=0.1652 |
val_SSIM=0.9631 | no improve 15/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:08<00:00, 84.5MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:09<00:00, 73.3MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 59/500 -> cminhnguyndsdssds/conditional-gan2d-normalized

Epoch 60/500 | loss_G=0.1654 | loss_D=0.5642 | loss_L1=0.1635 |
val_SSIM=0.9640 | no improve 16/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 90.9MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:09<00:00, 74.5MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 60/500 -> cminhnguyndsdssds/conditional-gan2d-normalized

Epoch 61/500 | loss_G=0.6787 | loss_D=0.5696 | loss_L1=0.1685 |
val_SSIM=0.9755 | no improve 17/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version

(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 88.4MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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100%|██████████| 662M/662M [00:06<00:00, 113MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 61/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 62/500 | loss_G=0.7300 | loss_D=0.5602 | loss_L1=0.1636 |
val_SSIM=0.9770 | no improve 18/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:05<00:00, 132MB/s]
2%| | 13.7M/662M [00:00<00:04, 144MB/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:08<00:00, 84.5MB/s]

Upload successful: last_checkpoint.pth (662MB)

Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdlds/conditional-gan2d-normalized>

Cloud: 62/500 -> cminhnguyndsdlds/conditional-gan2d-normalized
Epoch 63/500 | loss_G=0.7621 | loss_D=0.5571 | loss_L1=0.1614 |
val_SSIM=0.9558 | no improve 19/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 96.2MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

Upload successful: best_model.pth (662MB)

Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:05<00:00, 134MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 63/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Epoch 64/500 | loss_G=0.5856 | loss_D=0.5479 | loss_L1=0.1624 |
val_SSIM=0.9673 | no improve 20/20

Warning: Looks like you're using an outdated `kaggle` version
(installed: 2.0.1), please consider upgrading to the latest version
(2.0.2)

Starting upload for file best_model.pth

100%|██████████| 662M/662M [00:07<00:00, 91.0MB/s]
0%| | 0.00/662M [00:00<?, ?B/s]

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Starting upload for file last_checkpoint.pth

100%|██████████| 662M/662M [00:08<00:00, 86.6MB/s]

Upload successful: last_checkpoint.pth (662MB)
Dataset version is being created. Please check progress at
<https://www.kaggle.com/datasets/cminhnguyndsdssds/conditional-gan2d-normalized>

Cloud: 64/500 -> cminhnguyndsdssds/conditional-gan2d-normalized
Early stopping tai epoch 64

=== TRAINING HOAN THANH ===

Best val_SSIM: 0.9778

Model lưu tại:

/kaggle/working/conditional_gan2d_normalized/best_model.pth

Push da duoc tich hop vao training loop (cell tren) – chay moi epoch.

print(f'Checkpoints da duoc push len: {KAGGLE_USER}/{DATASET_NAME}')

Checkpoints da duoc push len: cminhnguyndsdssds/conditional-gan2d-normalized